

= Separation method:

An equation of the form:

$$F(x)G(y)dx + f(x)g(y)dy = 0$$

is called an equation with variable separable
or simple a separation equation.

→ dx x = dy y
→ integration

$$1) x \sin y \cdot dx + (x^2 + 1) \cos y \cdot dy = 0$$

$$2) (x-4)y^4 dx - x^3(y^2-3) dy = 0$$

$$3) (e^v + 1) \cos u \cdot du + e^v (\sin u + 1) \cdot dv = 0$$

$$4) (y+2) dx + y(x+4) dy = 0, y(-3) = -1$$

$$5) 8 \cos^2 y \cdot dx + \operatorname{cosec}^2 x \cdot dy = 0, y\left(\frac{\pi}{12}\right) = \frac{\pi}{4}$$

$$\frac{dy}{dx} = \frac{2+x}{y^2 x}$$

$$y^2 x \cdot \frac{dy}{dx} = 2+x$$

$$\frac{(1-y^2)^{-1/2} \cdot (-2y) \cdot dy}{dx} = \frac{2+x}{y^2 x}$$

$$(1) x \sin y dx + (x^2+1) \cos y dy = 0$$

$$\Rightarrow \frac{x \sin y dx}{\sin y (x^2+1)} + \frac{(x^2+1) \cos y dy}{\sin y (x^2+1)} = \frac{0}{\sin y (x^2+1)}$$

$$\Rightarrow \frac{x}{x^2+1} dx + \frac{\cos y}{\sin y} dy = 0$$

$$\Rightarrow \int \frac{2x}{x^2+1} dx + \int \frac{\cos y}{\sin y} dy = \int 0$$

$$\Rightarrow \frac{1}{2} \ln(x^2+1) + \ln|\sin y| = \ln|0|$$

$$\Rightarrow (x^2+1)^{1/2} \sin y = C$$

$$\Rightarrow (\sqrt{x^2+1}) \sin y = C$$

$$(2) (x-4) y^4 dx - x^3 (y^2-3) dy = 0$$

$$\Rightarrow \frac{x-4}{y^4 x^3} y^4 dx - \frac{x^3 (y^2-3)}{y^4 x^3} dy = 0$$

$$\Rightarrow \frac{x-4}{x^3} dx - \frac{y^2-3}{y^4} dy = 0$$

$$\Rightarrow \int \frac{x-4}{x^3} dx - \int \frac{y^2-3}{y^4} dy = \int 0$$

$$\Rightarrow \int \frac{x}{x^3} dx - \int \frac{4}{x^3} dx - \int \frac{y^2}{y^4} dy + \int \frac{3}{y^4} dy = \ln|0|$$

$$\Rightarrow \int \frac{1}{x^2} dx - \int 4x^{-3} dx - \int \frac{1}{y^2} dy + \int 3y^{-4} dy = C$$

$$\Rightarrow \frac{x^{-1}}{(-1)} - \frac{4x^{-2}}{(-2)} - \frac{y^{-1}}{(-1)} + \frac{3y^{-3}}{(-3)} = C$$

$$\Rightarrow -\frac{1}{x} + \frac{2}{x^2} + \frac{1}{y} - \frac{1}{y^3} = C$$

(A)

$$(3) (e^v+1) \cos u du + e^v (\sin u+1) dv = 0$$

$$\Rightarrow \frac{(e^v+1) \cos u}{(e^v+1)(\sin u+1)} du + \frac{e^v (\sin u+1)}{(e^v+1)(\sin u+1)} dv = 0$$

$$\Rightarrow \frac{\cos u}{\sin u+1} du + \frac{e^v}{e^v+1} dv = 0$$

$$\Rightarrow \int \frac{\cos u}{\sin u+1} du + \int \frac{e^v}{e^v+1} dv = \int 0$$

$$\Rightarrow \ln|\sin u+1| + \ln|e^v+1| = \ln|0|$$

$$\Rightarrow \ln|(\sin u+1)(e^v+1)| = \ln|0|$$

$$\Rightarrow (\sin u+1)(e^v+1) = C$$

(A)

$$\left[\int \frac{f'(x)}{f(x)} = \ln|f(x)| \right]$$

$$(4) (y+2)dx + y(x+4)dy = 0, \quad y(-3) = -1$$

$$\Rightarrow \frac{(y+2)}{(y+2)(x+4)} dx + y \frac{(x+4)}{(x+4)(y+2)} dy = 0$$

$$\Rightarrow \frac{1}{x+4} dx + \frac{y}{y+2} dy = 0$$

$$\Rightarrow \int \frac{1}{x+4} dx + \int \frac{y+2-2}{y+2} dy = \int 0$$

$$\Rightarrow \ln|x+4| + \int 1 \cdot \frac{y+2}{y+2} dy - \int \frac{2}{y+2} dy = 0 + C$$

$$\Rightarrow \ln|x+4| + y - 2 \cdot \ln|y+2| = C$$

$$\Rightarrow \ln|x+4| + y - \ln|y+2|^2 = C$$

$$\Rightarrow \ln \left\{ \frac{x+4}{(y+2)^2} \right\} + y = C$$

$$y(-3) = -1$$

$$\Rightarrow \ln \left(\frac{-3+4}{(-1)^2} \right) + (-1) = C$$

$$\Rightarrow \ln \left(\frac{1}{1} \right) - 1 = C$$

$$\Rightarrow C = -1$$

$$\boxed{\ln \left\{ \frac{x+4}{(y+2)^2} \right\} + y + 1 = 0} \quad (A)$$

$$(5) \quad 8 \cos^2 y \, dx + \operatorname{cosec}^2 x \, dy = 0, \quad y\left(\frac{\pi}{12}\right) = \frac{\pi}{4}$$

$$\Rightarrow 8 \frac{\cos^2 y}{\cos^2 y \operatorname{cosec}^2 x} dx + \frac{\operatorname{cosec}^2 x}{\cos^2 y \operatorname{cosec}^2 x} dy = 0$$

$$\Rightarrow 8 \frac{1}{\operatorname{cosec}^2 x} dx + \frac{1}{\cos^2 y} dy = 0 \quad \left[\frac{1}{\operatorname{cosec}^2 x} = \sin^2 x \right]$$

$$\Rightarrow 8 \int \sin^2 x \, dx + \int \sec^2 y \, dy = 0 \quad \left[\frac{1}{\cos^2} = \sec^2 \right]$$

$$\Rightarrow 8 \int \frac{1 - \cos 2x}{2} dx + \tan y = 0 \quad \left[2 \sin^2 x = 1 - \cos 2x \right]$$

$$\left[\tan y = \sec^2 y \right]$$

$$\Rightarrow 8 \int \frac{1}{2} - \frac{\cos 2x}{2} dx + \tan y = c$$

$$\Rightarrow 8 \left(\frac{1}{2} x - \frac{\sin 2x}{2 \cdot 2} \right) + \tan y = c$$

$$\Rightarrow 4x - 2 \sin 2x + \tan y = c$$

$$\therefore y\left(\frac{\pi}{12}\right) = \frac{\pi}{4}$$

$$\Rightarrow 4\left(\frac{\pi}{12}\right) - 2 \sin 2\left(\frac{\pi}{12}\right) + \tan\left(\frac{\pi}{4}\right) = c$$

$$\Rightarrow \frac{\pi}{3} - 2 \sin\left(\frac{\pi}{6}\right) + \tan(45) = c$$

$$\Rightarrow \frac{\pi}{3} = c$$

$$4x - 2 \sin 2x + \tan y = \frac{\pi}{3}$$

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